KisanAI: Deep Learning-Based Crop Disease Detection System for Indian Farmers Using Transfer Learning and Grad-CAM Visualization

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GitHub: github.com/soveshika/kisanai

Abstract

Crop disease is one of the leading causes of agricultural loss in India, with farmers losing an estimated 20-30% of their annual yield due to undetected or misdiagnosed plant diseases. Early and accurate detection is critical, yet most smallholder farmers lack access to agricultural experts. This paper presents KisanAI, a deep learning system for automated crop disease detection that supports 29 disease classes across six major Indian crops: tomato, potato, bell pepper, rice, wheat, and cotton. The system employs EfficientNetB0 with transfer learning, achieving 89% validation accuracy on a combined dataset of 23,994 images. Gradient-weighted Class Activation Mapping (Grad-CAM) provides visual explanations of model predictions, highlighting the diseased regions on the leaf. Treatment recommendations are delivered in both Hindi and English, making the system accessible to Indian farmers. The complete system is deployed as a Streamlit web application and is publicly available on GitHub.

Keywords: Plant Disease Detection, Transfer Learning, EfficientNet, Grad-CAM, Deep Learning, Agricultural AI, Precision Farming

1. Introduction

India is an agrarian economy with over 58% of its rural population dependent on agriculture for their livelihood. Plant diseases pose a severe threat to food security, causing significant economic losses to farmers every year. Traditional methods of disease detection rely on manual inspection by agricultural experts, which is both time-consuming and inaccessible to most smallholder farmers, particularly in rural areas of states like Haryana, Punjab, and Uttar Pradesh.

The rapid advancement of deep learning and computer vision has opened new possibilities for automated plant disease detection. Convolutional Neural Networks (CNNs) have demonstrated remarkable performance in image classification tasks, including agricultural applications. However, most existing systems are limited to a small number of crops or require expensive computational infrastructure.

This paper presents KisanAI, a comprehensive crop disease detection system that addresses these limitations. KisanAI supports 29 disease classes across six major crops, provides bilingual output in Hindi and English, and includes Grad-CAM visualizations to help farmers and agricultural officers understand the model's decision-making process. The system is designed to be accessible via a simple web interface, requiring no technical expertise from the end user.

2. Related Work

Mohanty et al. (2016) demonstrated that deep learning could achieve over 99% accuracy on the PlantVillage dataset using a CNN trained on 54,306 images across 26 disease classes. However, their model was trained on controlled laboratory images and showed reduced performance on field conditions. Ferentinos (2018) used AlexNet and GoogLeNet architectures and achieved 99.53% accuracy, though again limited to controlled conditions.

More recent work has focused on transfer learning approaches. Karthik et al. (2020) applied ResNet50 with transfer learning for tomato disease detection, achieving 91.2% accuracy. EfficientNet, introduced by Tan and Le (2019), offers superior performance with fewer parameters through compound scaling, making it suitable for resource-constrained deployment.

Explainability in agricultural AI has received growing attention. Grad-CAM, proposed by Selvaraju et al. (2017), produces visual explanations by highlighting discriminative regions in the input image. Its application in plant disease detection helps build trust with farmers and agricultural officers by showing which leaf regions triggered the diagnosis.

KisanAI extends previous work by combining multiple crop datasets, applying transfer learning with fine-tuning, incorporating Grad-CAM explainability, and delivering bilingual output specifically targeting Indian farming communities.

3. Dataset

The training dataset was assembled from four publicly available sources, combined into a unified dataset of 29,979 images across 29 classes. An 80/20 train-validation split was applied, yielding 23,994 training images and 5,985 validation images.

Source	Crops	Classes	Images
PlantVillage	Tomato, Potato, Bell Pepper	15	~20,000
Rice Disease Dataset	Rice	6	3,829
Cotton Disease Dataset	Cotton	4	2,000+
Wheat Disease Dataset	Wheat	4	4,000+
Combined Total	**6 Crops**	**29**	**29,979**

Table 1: Dataset composition by source, crop, class count, and image count.

Data augmentation was applied during training to improve generalisation and reduce overfitting. Augmentation techniques included random rotation (up to 20 degrees), horizontal flipping, width and height shifts (up to 20%), and EfficientNet-specific preprocessing using the preprocess_input function from TensorFlow Keras.

4. Methodology

4.1 Model Architecture

EfficientNetB0 was selected as the base model due to its superior accuracy-to-parameter ratio compared to VGG, ResNet, and InceptionNet architectures. The model was pre-trained on ImageNet (14 million images, 1,000 classes), providing rich feature representations transferable to plant disease classification.

The final architecture consists of: (1) EfficientNetB0 base with frozen weights, (2) Global Average Pooling layer to reduce spatial dimensions, (3) Dropout layer (rate=0.3) for regularisation, (4) Dense layer with 256 units and ReLU activation, (5) second Dropout layer (rate=0.3), and (6) output Dense layer with 29 units and Softmax activation.

4.2 Training Strategy

Training was conducted in two phases. In Phase 1, the EfficientNetB0 base layers were frozen and only the newly added top layers were trained for 10 epochs using the Adam optimiser with default learning rate (1e-3) and categorical cross-entropy loss. This allowed the top layers to converge before fine-tuning.

In Phase 2, the top 20 layers of EfficientNetB0 were unfrozen and the entire network was fine-tuned for an additional 10 epochs using a reduced learning rate (1e-5) to prevent catastrophic forgetting of the pre-trained features. Training was performed on Google Colab with NVIDIA T4 GPU acceleration.

4.3 Grad-CAM Explainability

Gradient-weighted Class Activation Mapping (Grad-CAM) was implemented to provide visual explanations of model predictions. The technique computes the gradient of the predicted class score with respect to the feature maps of the final convolutional layer (top_conv in EfficientNetB0). These gradients are globally average pooled to produce weights, which are then used to create a weighted combination of the feature maps, resulting in a heatmap highlighting the discriminative regions.

The heatmap is superimposed on the original image using a jet colormap at 40% opacity, making it possible to visually verify that the model is focusing on the diseased regions of the leaf rather than background artifacts.

5. Results

The model achieved 89% validation accuracy after Phase 1 training. The two-phase training strategy proved effective, with Phase 1 establishing a strong baseline and Phase 2 fine-tuning providing additional improvements on difficult classes.

Metric	Value
Training Images	23,994
Validation Images	5,985
Number of Classes	29
Validation Accuracy	89%
Model Parameters	~4.2M (EfficientNetB0)
Input Resolution	224 x 224 RGB
Training Platform	Google Colab T4 GPU

Table 2: Summary of model performance metrics.

Single image inference time was approximately 0.5 seconds on CPU (Apple M4), making the system suitable for real-time use in the web application. Grad-CAM visualisations consistently highlighted the diseased leaf regions, confirming that the model learned meaningful disease-related features rather than background patterns.

6. System Implementation

KisanAI is implemented as a modular Python application with four core components. The prediction module (predict.py) handles image preprocessing and model inference. The treatment advice module (treatment_advice.py) maps disease predictions to structured treatment recommendations in Hindi and English. The Grad-CAM module (gradcam.py) generates visual explanations for each prediction. The web application (app.py) integrates all components into a Streamlit interface.

The treatment advice database covers all 29 disease classes with chemical treatment options, organic alternatives, and prevention strategies. Hindi translations ensure accessibility for farmers with limited English proficiency. The web application accepts uploaded leaf images and returns the disease diagnosis, confidence score, Grad-CAM visualisation, and complete treatment advice within seconds.

7. Conclusion

This paper presented KisanAI, a deep learning system for crop disease detection targeting Indian farmers. The system achieves 89% validation accuracy across 29 disease classes using EfficientNetB0 with transfer learning. Grad-CAM visualisations provide interpretable predictions, and bilingual output in Hindi and English ensures accessibility to the target user base.

Future work will focus on expanding the crop coverage to include maize, sugarcane, and mustard diseases. WhatsApp bot integration is planned to enable farmers to receive diagnoses without requiring internet browser access. Mobile app deployment and voice output in regional Indian languages including Punjabi and Gujarati are also under consideration. The authors also plan to collect real field data from Indian farms to improve model robustness under uncontrolled lighting and environmental conditions.

8. References

[1] Mohanty, S. P., Hughes, D. P., & Salathe, M. (2016). Using deep learning for image-based plant disease detection. Frontiers in Plant Science, 7, 1419.

[2] Ferentinos, K. P. (2018). Deep learning models for plant disease detection and diagnosis. Computers and Electronics in Agriculture, 145, 311-318.

[3] Tan, M., & Le, Q. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. International Conference on Machine Learning (ICML).

[4] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. ICCV.

[5] Karthik, R., Hariharan, M., Anand, S., Mathikshara, P., Johnson, A., & Menaka, R. (2020). Attention embedded residual CNN for disease detection in tomato leaves. Applied Soft Computing, 86, 105933.

[6] Hughes, D., & Salathe, M. (2015). An open access repository of images on plant health to enable the development of mobile disease diagnostics. arXiv:1511.08060.